

# Assignment 1: Git and Python

## Software Engineering for Scientists

**Objectives:** Become familiar with Python.

**Tasks:**

1. Fork the repo <https://github.com/ryla3858/python-refresher>
2. Clone **your** fork (NOT <https://github.com/swe4s/python-refresher>)
3. In the root directory of the repository (i.e., after you `cd python_refresher`), add the agricultural CO2 emission data file called `Agrofood_co2_emission.csv` in the Assignments > Data Google Drive folder

### DO NOT COMMIT THIS FILE TO YOUR REPO

4. Make a branch
5. Complete the implementation of `get_column()` in `my_utils.py` such that
  - a. `get_column()` opens the file named `file_name` and processes it line by line
  - b. for each line
    - i. split the line into an array
    - ii. check to see if the value in the `query_column` position of the array matches the value stored in the `query_value` variable
    - iii. when the above condition is met, add the value in the `result_column` position to an array
  - c. return the array storing the column values
6. Commit changes to the branch with useful commit messages
7. Update `print_fires.py` to correctly use `get_column()` to print the emissions due to fires in United States of America (or any other country of your choice)
8. Commit changes to the branch with useful commit messages
9. Improve `get_column()` to use a named argument for the `result_column` variable that defaults to 1
10. Update `print_fires.py` to correctly use new `get_column()`
11. Create a new file `run.sh` that runs `print_fires.py`
12. Update `README.md` to summarize your changes
13. Commit changes to the branch with useful commit messages
14. Push the branch to your repository
15. Create a pull request from that branch (**SEE BELOW**)
16. Merge the pull request
17. Create a release from master tagged as v1.0

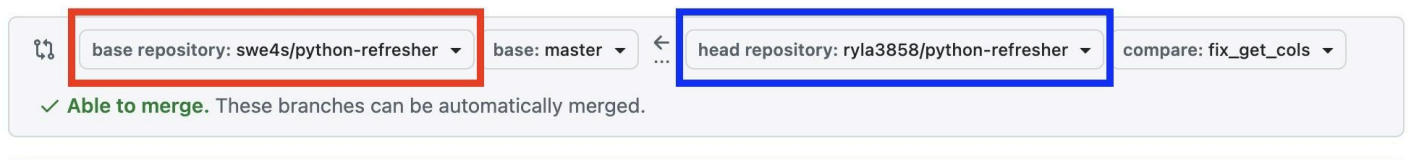
# Creating a pull request to forked repo

A pull request has a **base repo** and a **head repo**. The base repo is the target you're merging into, and the head repo is the source containing your changes. When you make a pull request from a forked repo, the **head repo is your fork** (the repository you pushed your branch to from your local environment). By default, GitHub sets the **base repo to the original repo you forked from**. This usually makes sense because pull requests are normally contributing changes back to the upstream project. But for our homework, you want to merge changes within your own fork's own main branch, and you need to change the base repo to your forked repo instead of leaving it as the original.

Default:

## Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

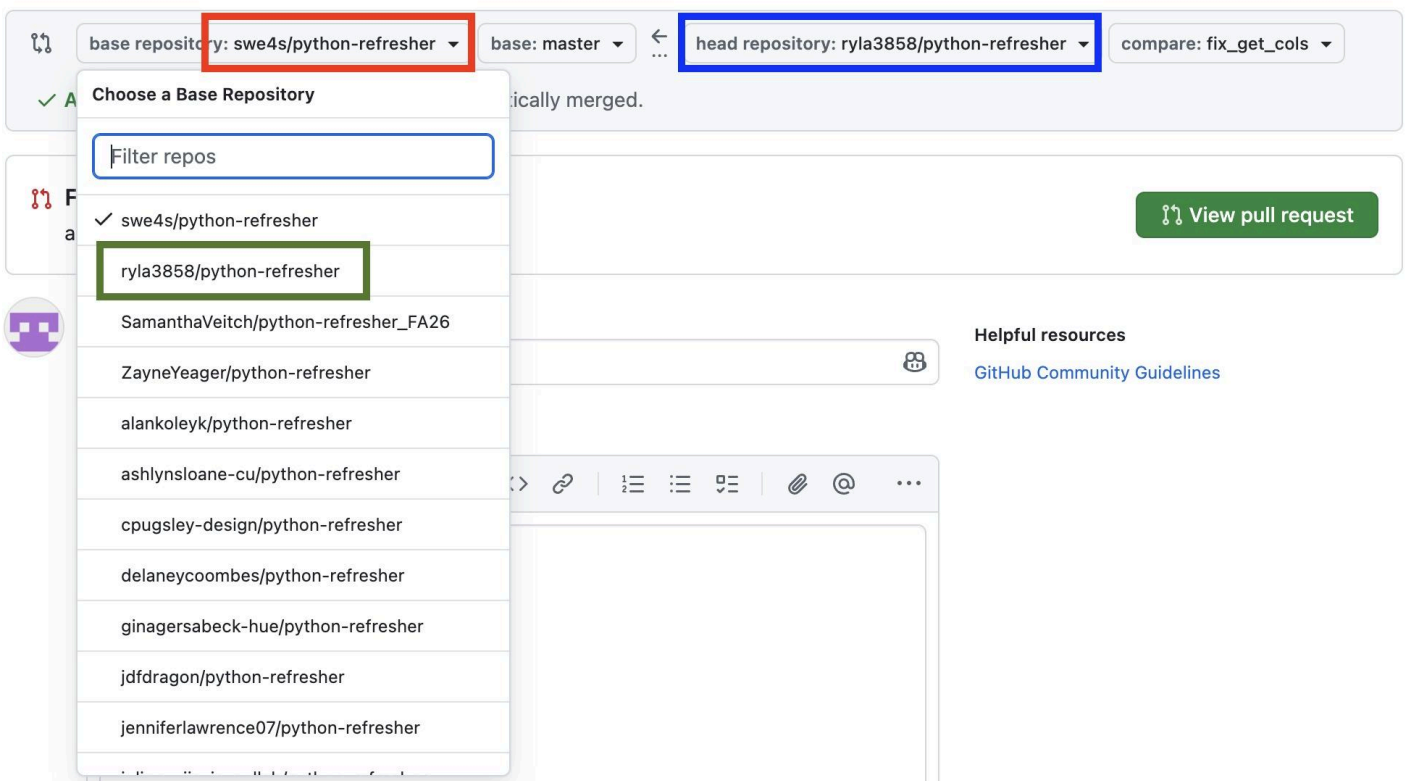


The screenshot shows the GitHub interface for creating a pull request. The 'base repository' dropdown is set to 'swe4s/python-refresher' and is highlighted with a red box. The 'base' branch is 'master'. The 'head repository' dropdown is set to 'ryla3858/python-refresher' and is highlighted with a blue box. The 'compare' dropdown is set to 'fix\_get\_cols'. A green checkmark and the text 'Able to merge. These branches can be automatically merged.' are visible below the repository selection.

Pick the main (master here) branch as the base repo:

## Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).



The screenshot shows the GitHub interface with the 'base repository' dropdown menu open. The dropdown is highlighted with a red box. The menu lists several repositories, with 'ryla3858/python-refresher' highlighted by a green box. The 'head repository' dropdown is set to 'ryla3858/python-refresher' and is highlighted with a blue box. The 'compare' dropdown is set to 'fix\_get\_cols'. A green checkmark and the text 'Able to merge. These branches can be automatically merged.' are visible below the repository selection. A green button labeled 'View pull request' is visible on the right side of the form.

Corrected base repo

## Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).



base: master ▾

← ...

compare: fix\_get\_cols ▾

✓ **Able to merge.** These branches can be automatically merged.